

**BONANG SEPHEU**

**ST10443731**

**PROGRAMMING ASSIGNMENT PART 1**

**RACEDAY-API ENDPOINT PLAN**

## AUTHENTICATION

| HTTP Method | Route              | Description                                                      | Role Required | Required Body                  | Expected Response                                                                                                 |
|-------------|--------------------|------------------------------------------------------------------|---------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Post        | /api/auth/register | Creates a new user account as either an organizer or participant | None(public)  | {fullName,email,password,role} | 201 Created-returns new user id and role.400 Bad Request validation failed.409 conflict email already registered. |
| Post        | /api/auth/login    | Authenticates a user and returns a JWT for subsequent requests.  | None(public)  | { email, password }            | 200 OK-returns JWT token and role.401 Unauthorized-invalid credentials.                                           |

## USER PROFILE

| HTTP Method | Route         | Description                                              | Role Required | Request Body        | Expected Response                                          |
|-------------|---------------|----------------------------------------------------------|---------------|---------------------|------------------------------------------------------------|
| GET         | /api/users/me | Returns the profile of the currently logged in user.     | Any           | None                | 200 Ok-Returns user profile 401 Unauthorized-invalid token |
| PUT         | /api/users/me | Updates the profile details of the currently logged-user | Any           | { fullName, email } | 200 OK-updated profile. 400 Bad request-Validation failed  |

## EVENTS

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response |
|-------------|-------|-------------|---------------|--------------|-------------------|
|-------------|-------|-------------|---------------|--------------|-------------------|

|        |                                                            |                                                                             |           |                                                    |                                                                                  |
|--------|------------------------------------------------------------|-----------------------------------------------------------------------------|-----------|----------------------------------------------------|----------------------------------------------------------------------------------|
| Get    | /api/events                                                | Lists all upcoming events for participants to browse                        | Any       | None                                               | 200 OK-returns array of events                                                   |
| Get    | /api/events/{id}                                           | Returns full details full detail for single event, including its categories | Any       | None                                               | 200 OK – returns event detail. 404 Not Found – event does not exist.             |
| POST   | /api/events                                                | Creates a new event owned by the logged in organizer.                       | Organizer | { name,description, eventDate, location,routeInfo} | 201 Created-returns new event id. 400 Bad Request-validation failed.             |
| PUT    | Updates an existing event owned by the logged in organizer | Updates an existing event owned by the logged in organizer                  | Organizer | {name, description, eventDate, location,routeInfo} | 200 OK-returns updated event. 403 Forbidden – not the event owner.404 Not Found. |
| DELETE | /api/events/{id}                                           | Deletes an event owned by the logged in organizer.                          | Organizer | None                                               | 204 No Content. 403 Forbidden – not the event owner. 404 Not Found.              |
|        |                                                            |                                                                             |           |                                                    |                                                                                  |

## CATEGORIES

| HTTP Method | Route                            | Description                               | Role Required | Request Body        | Expected Response                                                         |
|-------------|----------------------------------|-------------------------------------------|---------------|---------------------|---------------------------------------------------------------------------|
| GET         | /api/events/{eventId}/categories | Lists all categories for a specific event | Any           | None                | 200 OK-returns array of categories<br>404 Not Found-event does not exist. |
| POST        | /api/events/{eventId}/categories | Adds a new                                | Organizer     | { name, distancekm, | 201 Created – returns new                                                 |

|        |                      |                                         |           |                                          |                                                                           |
|--------|----------------------|-----------------------------------------|-----------|------------------------------------------|---------------------------------------------------------------------------|
|        |                      | category to an event.                   |           | maxParticipants<br>}                     | category id.<br>403<br>Forbidden<br>not the event owner.404<br>Not Found. |
| PUT    | /api/categories/{id} | Updates an existing category's details. | Organizer | [ name, distancekm, maxParticipants<br>} | 200 OK-returns Updated category.403<br>Forbidden.404<br>Not Found.        |
| DELETE | /api/categories/{id} | Removes a category from an event.       | Organizer | None                                     | 204 No content.403<br>Forbidden.404<br>Not Found.                         |

### Event Enrolments

| HTTP Method | Route                                     | Description                                               | Role Required | Request Body | Expected Response                                                                                                                        |
|-------------|-------------------------------------------|-----------------------------------------------------------|---------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Post        | /api/categories/{categoriesId}/enrolments | Enrols the logged in participants into, if space remains. | Participant   | None         | 201 Created-returns enrolment id and status.<br>404 Not found-category does not exist.409<br>Conflict-already enrolled or category full. |
| GET         | /api/users/me/enrolments                  | Lists all enrolments for the logged in participant.       | participant   | None         | 200 OK-returns array of enrolments.                                                                                                      |
| GET         | /api/events/{eventId}/enrolments          | List all enrolments for an event, for the                 | Organiser     | None         | 200 OK – returns of enrolment                                                                                                            |

|        |                     |                                                 |             |      |                                                                         |
|--------|---------------------|-------------------------------------------------|-------------|------|-------------------------------------------------------------------------|
|        |                     | organizer to manage                             |             |      | s. 403 Forbidden – not found- Not Found.                                |
| DELETE | /api/enrolment/{id} | Cancels/withdraws a participant's own enrolment | participant | None | 204 No Content. 403 Forbidden – not the enrolment owner. 404 Not Found. |

## Results

| HTTP Method | Route                                 | Description                                                                    | Role Required | Request Body              | Expected Response                                                                                                                               |
|-------------|---------------------------------------|--------------------------------------------------------------------------------|---------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| POST        | /api/enrolments/{enrolmentsId}/result | Captures the finish time and position for a participant's enrolment            | Organizer     | { finishTime , position } | 201 Created-returns new result. 403 Forbidden - not the event. 404 Not Found-enrollment does not exist. 409 Conflict – result already captured. |
| GET         | /api/users/me/results                 | Returns the logged in participant's personal result history across all events. | Participant   | None                      | 200 OK-returns array of results.                                                                                                                |

|     |                                |                               |     |      |                                                                      |
|-----|--------------------------------|-------------------------------|-----|------|----------------------------------------------------------------------|
| GET | /api/events/{eventId}/ results | Returns all results for event | Any | None | 200 OK-returns array of result. 404 Not Found – event does not exist |
|-----|--------------------------------|-------------------------------|-----|------|----------------------------------------------------------------------|